Worksheet: 1.1–1.6 and 5.1–5.3

Name: _____ Date: _____

One Easy (E) and one Hard (H) problem per section.

1.1 Change in tandem

1E. Use the table. Is f increasing or decreasing? Concave up or concave down? Justify with the changes.

x	0	1	2	3	4
$f(x)$	10	6	3	1	0

1H. A function is **decreasing and concave down** on $(0, 3)$, then **decreasing and concave up** on $(3, 6)$.

- (a) Sketch a graph that does this.
- (b) What is special about $x = 3$, and what is the rate of change doing there?
- (c) Give a real-world situation that behaves like this.

1.2 Rates of change

2E. Find the average rate of change of $f(x) = x^2 + 3x$ on $[1, 4]$.

2H. A car's position is $s(t) = t^3$ feet after t seconds.

- (a) Find the average velocity on $[2, 2.1]$ and on $[1.9, 2]$. (Calculator OK)
- (b) Use them to estimate the velocity at exactly $t = 2$. Include units.

1.3 Linear and quadratic functions

3E. Linear, quadratic or neither? Write the function.

x	0	1	2	3
y	5	8	11	14

3H. A quadratic f has $f(0) = 3$, $f(1) = 6$, $f(2) = 13$.

- (a) Use first and second differences to predict $f(3)$ and $f(4)$.
- (b) Find the formula $f(x) = ax^2 + bx + c$ and check your predictions.

1.4 Polynomials and rates of change

4E. $f(x) = (x+3)(x-1)(x-4)$. List the zeros. Is f positive or negative on $(1, 4)$?

4H. Write a degree-4 polynomial with a **double zero at** $x = -1$, a **double zero at** $x = 3$, and $f(0) = 18$.

- (a) Find it in factored form.
- (b) What does the graph do at each zero? What is the end behavior?
- (c) What is the most turning points it can have?

1.5 Complex zeros, multiplicity, symmetry

5E. Solve $x^2 - 6x + 13 = 0$.

5H. $f(x) = x^3 + 2x^2 + 9x + 18$.

- (a) Find **all** zeros, real and complex. (Hint: grouping.)
- (b) How many times does the graph touch the x-axis? Why so few?

1.6 End behavior

6E. Describe the end behavior of $f(x) = -3x^4 + x$ in words and with limit notation.

6H. $f(x) = -2(x-1)^3(x+2)^2$.

- (a) Find the leading term without multiplying everything out, and give the end behavior.
- (b) Describe the graph at each zero, and find the y-intercept.
- (c) Sketch it.

5.1 Composition

7E. $f(x) = 3x - 1$ and $g(x) = x^2 + 2$. Find $f(g(2))$ and $g(f(2))$.

7H. $f(x) = \sqrt{x-1}$ and $g(x) = \dfrac{1}{x-3}$.

- (a) Find $f(g(x))$.
- (b) Find its domain. (Both steps must work: g has to be defined, and $g(x)$ has to be allowed inside f.)

5.2 Inverses

8E. Find the inverse of $f(x) = 4x + 7$, and check that $f(f^{-1}(x)) = x$.

8H. $f(x) = \dfrac{2x-1}{x+3}$.

- (a) Find $f^{-1}(x)$.
- (b) Give the domain of f and of f^{-1}.
- (c) $f(1) = \frac{1}{4}$. Use your inverse to confirm $f^{-1}\left(\frac{1}{4}\right) = 1$.

5.3 Exponential functions

9E. Find the exponential function $y = ab^x$ through $(0, 5)$ and $(1, 15)$. Is it growth or decay, and by what percent?

9H.

- (a) Solve $4^{x+1} = 8^x$ by writing both sides with the same base.
- (b) A bacteria culture starts with 500 and **triples every 4 hours**. Write a model, find the population after 10 hours (calculator OK), and find exactly when it reaches 4500.

Answer Key

1E. Changes: $-4, -3, -2, -1$. All negative, so **decreasing**. The changes are increasing (getting less negative), so the rate of change is increasing: **concave up**.

1H. (a) Falling and curving down, then still falling but flattening out. (b) $x = 3$ is a **point of inflection**. The rate of change is decreasing up to $x = 3$ and increasing after, so the graph is falling **fastest** at $x = 3$ (the most negative rate). (c) A sled going down a hill that's steepest in the middle; or a tank draining faster and faster, then easing off.

2E. $f(4) = 28$, $f(1) = 4$, so $\dfrac{28-4}{4-1} = \boxed{8}$.

2H. (a) $\dfrac{2.1^3 - 8}{0.1} = \dfrac{9.261 - 8}{0.1} = 12.61$ ft/s and $\dfrac{8 - 1.9^3}{0.1} = \dfrac{8 - 6.859}{0.1} = 11.41$ ft/s. (b) The true value is between them, about **12 ft/s** (their average is 12.01).

3E. Differences all 3, so **linear**: $y = 3x + 5$.

3H. (a) First differences $3, 7$; second difference 4. Continue: next first differences $11, 15$, so $f(3) = 13 + 11 = 24$ and $f(4) = 24 + 15 = 39$. (b) $c = 3$; $a + b = 3$; $4a + 2b = 10$, so $a = 2$, $b = 1$: $f(x) = 2x^2 + x + 3$. Check: $f(3) = 18 + 3 + 3 = 24$ ✓, $f(4) = 32 + 4 + 3 = 39$ ✓.

4E. Zeros $-3, 1, 4$. Test $x = 2$: $(5)(1)(-2) = -10$, so **negative** on $(1, 4)$.

4H. (a) $f(x) = a(x+1)^2(x-3)^2$ and $f(0) = a(1)(9) = 9a = 18$, so $a = 2$: $\boxed{f(x) = 2(x+1)^2(x-3)^2}$. (b) Both zeros have even multiplicity, so the graph **bounces** at -1 and at 3. The leading term is $2x^4$: up on both ends. (c) At most $4 - 1 = 3$ turning points (it has exactly 3: min, max, min).

5E. $x = \dfrac{6 \pm \sqrt{36 - 52}}{2} = \dfrac{6 \pm 4i}{2} = \boxed{3 \pm 2i}$

5H. (a) $x^2(x+2) + 9(x+2) = (x+2)(x^2+9)$, so $x = -2, \pm 3i$. (b) Only **once**, at $x = -2$. The other two zeros are complex, so the graph never reaches them.

6E. Leading term $-3x^4$: even degree, negative, so **down on both ends**. $\lim_{x \to \pm\infty} f(x) = -\infty$.

6H. (a) $-2 \cdot x^3 \cdot x^2 = -2x^5$: odd degree, negative. $\lim_{x \to -\infty} f = \infty$ and $\lim_{x \to \infty} f = -\infty$. (b) At $x = 1$, multiplicity 3: **crosses**, flattening like x^3. At $x = -2$, multiplicity 2: **bounces**. y-intercept: $-2(-1)^3(2)^2 = -2(-1)(4) = 8$. (c) Comes down from the top left, bounces off -2, rises through $(0, 8)$, turns, flattens through 1, then heads down.

7E. $g(2) = 6$, so $f(6) = 17$. $f(2) = 5$, so $g(5) = 27$.

7H. (a) $f(g(x)) = \sqrt{\dfrac{1}{x-3} - 1}$. (b) g needs $x \neq 3$. f needs its input ≥ 1: $\dfrac{1}{x-3} \geq 1$. That only works when $x - 3$ is positive and at most 1, so $0 < x - 3 \leq 1$. Domain: $\boxed{3 < x \leq 4}$.

8E. $x = 4y + 7$, so $y = \dfrac{x-7}{4}$. Check: $4 \cdot \dfrac{x-7}{4} + 7 = x$ ✓.

8H. (a) Swap: $x = \dfrac{2y-1}{y+3}$, so $xy + 3x = 2y - 1$, then $xy - 2y = -3x - 1$, then $y(x-2) = -3x - 1$. So $\boxed{f^{-1}(x) = \dfrac{3x+1}{2-x}}$. (b) Domain of f: $x \neq -3$. Domain of f^{-1}: $x \neq 2$ (which is the range of f). (c) $f^{-1}\left(\frac{1}{4}\right) = \dfrac{3/4 + 1}{2 - 1/4} = \dfrac{7/4}{7/4} = 1$ ✓.

9E. $a = 5$; $5b = 15$, so $b = 3$: $y = 5 \cdot 3^x$. Growth, $b - 1 = 2$, so **200%** per step.

9H. (a) $2^{2(x+1)} = 2^{3x}$, so $2x + 2 = 3x$ and $\boxed{x = 2}$. (b) $N(t) = 500 \cdot 3^{t/4}$. $N(10) = 500 \cdot 3^{2.5} \approx \boxed{7794}$. $500 \cdot 3^{t/4} = 4500$ gives $3^{t/4} = 9 = 3^2$, so $t/4 = 2$ and $\boxed{t = 8 \text{ hours}}$.